

EXTENDS *Naturals*

This model keeps only the states needed to visualize three delayed operations: a transfer confirmation, an old *G1* page logout, and old-cookie cleanup. *G3A* represents any successor installed while one of those operations is delayed.

RunOldPageCleanup treats inspection and mutation as one action. This is a protocol model under the implementation's enforced browser-profile Web Lock, which spans command creation, both *HTTP* responses, and response-side cleanup. It does not claim that the cleanup would be safe without that serialization.

Sessions $\triangleq$ {"G1A", "G2B", "G3A"}
Cookies $\triangleq$ {"G1", "G2", "G3", "none"}
ChallengeStates $\triangleq$ {"none", "pending", "used"}

VARIABLES

currentSession,
epoch,
cookieA,
cookieB,
challengeState,
successorInstalled,
revoked,
oldLogoutHandled,
oldCleanupHandled

vars $\triangleq$ $\langle$ *currentSession*, *epoch*, *cookieA*, *cookieB*, *challengeState*,
successorInstalled, *revoked*, *oldLogoutHandled*,
oldCleanupHandled $\rangle$

Init $\triangleq$

$\wedge$ *currentSession* = "G1A"
 $\wedge$ *epoch* = 1
 $\wedge$ *cookieA* = "G1"
 $\wedge$ *cookieB* = "none"
 $\wedge$ *challengeState* = "none"
 $\wedge$ *successorInstalled* = FALSE
 $\wedge$ *revoked* = FALSE
 $\wedge$ *oldLogoutHandled* = FALSE
 $\wedge$ *oldCleanupHandled* = FALSE

StageTransfer $\triangleq$

$\wedge$ *challengeState* = "none"
 $\wedge$ *challengeState'* = "pending"
 $\wedge$ UNCHANGED $\langle$ *currentSession*, *epoch*, *cookieA*, *cookieB*,
successorInstalled, *revoked*, *oldLogoutHandled*,
oldCleanupHandled $\rangle$

oldCleanupHandled⟩

ConfirmTransfer $\triangleq$
 $\wedge$ *challengeState* = “pending”
 $\wedge$ *challengeState'* = “used”
 $\wedge$ IF *currentSession* = “G1A” $\wedge$ *epoch* = 1 $\wedge$ $\neg$ *revoked*
 THEN $\wedge$ *currentSession'* = “G2B”
 $\wedge$ *epoch'* = 2
 $\wedge$ *cookieB'* = “G2”
 $\wedge$ UNCHANGED ⟨*cookieA*, *revoked*⟩
 ELSE $\wedge$ UNCHANGED ⟨*currentSession*, *epoch*, *cookieA*, *cookieB*,
 revoked⟩
 $\wedge$ UNCHANGED ⟨*successorInstalled*, *oldLogoutHandled*,
oldCleanupHandled⟩

InstallCompetingSuccessor $\triangleq$
 $\wedge$ *challengeState* $\neq$ “none”
 $\wedge$ $\neg$ *revoked*
 $\wedge$ *currentSession* $\neq$ “G3A”
 $\wedge$ *currentSession'* = “G3A”
 $\wedge$ *epoch'* = *epoch* + 1
 $\wedge$ *cookieA'* = “G3”
 $\wedge$ *successorInstalled'* = TRUE
 $\wedge$ UNCHANGED ⟨*cookieB*, *challengeState*, *revoked*,
oldLogoutHandled, *oldCleanupHandled*⟩

RunOldPageLogout $\triangleq$
 $\wedge$ *currentSession* $\neq$ “G1A”
 $\wedge$ $\neg$ *oldLogoutHandled*
 $\wedge$ *oldLogoutHandled'* = TRUE
 $\wedge$ *revoked'* = IF *currentSession* = “G1A” THEN TRUE ELSE *revoked*
 $\wedge$ UNCHANGED ⟨*currentSession*, *epoch*, *cookieA*, *cookieB*,
challengeState, *successorInstalled*,
oldCleanupHandled⟩

RunOldPageCleanup $\triangleq$
 $\wedge$ *currentSession* $\neq$ “G1A”
 $\wedge$ $\neg$ *oldCleanupHandled*
 $\wedge$ *oldCleanupHandled'* = TRUE
 $\wedge$ *cookieA'* = IF *cookieA* = “G1” THEN “none” ELSE *cookieA*
 $\wedge$ UNCHANGED ⟨*currentSession*, *epoch*, *cookieB*, *challengeState*,
successorInstalled, *revoked*, *oldLogoutHandled*⟩

Next $\triangleq$
 $\vee$ *StageTransfer*
 $\vee$ *ConfirmTransfer*

$\vee$ *InstallCompetingSuccessor*
 $\vee$ *RunOldPageLogout*
 $\vee$ *RunOldPageCleanup*

Spec $\triangleq$ *Init* $\wedge$ $\Box[Next]_{vars}$

TypeOK $\triangleq$
 $\wedge$ *currentSession* $\in$ *Sessions*
 $\wedge$ *epoch* $\in$ 1 .. 3
 $\wedge$ *cookieA* $\in$ *Cookies*
 $\wedge$ *cookieB* $\in$ *Cookies*
 $\wedge$ *challengeState* $\in$ *ChallengeStates*
 $\wedge$ *successorInstalled* $\in$ BOOLEAN
 $\wedge$ *revoked* $\in$ BOOLEAN
 $\wedge$ *oldLogoutHandled* $\in$ BOOLEAN
 $\wedge$ *oldCleanupHandled* $\in$ BOOLEAN

LiveSessionKeepsItsCookie $\triangleq$
 $\wedge$ (*currentSession* = "G1A" $\wedge$ $\neg$ *revoked*) $\Rightarrow$ *cookieA* = "G1"
 $\wedge$ *currentSession* = "G2B" $\Rightarrow$ *cookieB* = "G2"
 $\wedge$ *currentSession* = "G3A" $\Rightarrow$ *cookieA* = "G3"

OldPageCannotRevokeSuccessor $\triangleq$
currentSession $\neq$ "G1A" $\Rightarrow$ $\neg$ *revoked*

InstalledSuccessorCannotBeReplaced $\triangleq$
successorInstalled $\Rightarrow$ *currentSession* = "G3A"
